INCH POUND

W-B-133E
24 February 2005
SUPERSEDING
W-B-133D
26 July 1985

FEDERAL SPECIFICATION

BATTERY, STORAGE, LEAD-ACID (INDUSTRIAL MOTIVE POWER SERVICE)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers industrial lead-acid batteries for use in electric fork trucks, electric tractors, and other comparable motive power applications.

1.2 Classification. Storage batteries covered by this specification are classified by voltage, capacity (ampere-hours) and maximum dimensions.

1.2.1 Voltage. The voltage for the batteries covered by this specification shall be 12, 18, 24, 36, 48, 72, or 80 volts. The battery voltage shall be as specified in the acquisition order (see 6.2).

1.2.2 Capacity. The capacity of batteries covered by this specification shall be in the range from approximately 200 to approximately 2,000 ampere-hours. The battery capacity shall be as specified in the acquisition order (see 6.2).

1.2.3 Dimensions. The maximum length, width, and height dimensions for batteries covered by this specification shall be specified in the acquisition order to the tenth of an inch (see 6.2).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 6140

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

O-S-801 - Sulfuric Acid, Electrolyte (for Storage Batteries).

(Activities outside the federal government may obtain copies of federal specifications, standards, and commercial item descriptions as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index is for sale on a subscription basis from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Single copies of this specification, and other federal specifications and commercial item descriptions required by activities outside the federal government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Federal government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product.

(Copies of military standards required by contractors in connection with specific procurement functions are obtained from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. Electronic copies of standards may be obtained from http://assist.daps.dla.mil.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Battery Council International (BCI)

BCIS-10 - Recommended Practices for: Warning Messages, General Labeling & Marking, and Shipping & Packaging for North America (NAFTA Countries).

BCI-I-2 - Determination of Capacity of Lead-Acid Industrial Storage Batteries for Motive Power Service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

BCI-I-3 - Life Cycle Testing of Lead-Acid Industrial Storage Batteries for Motive Power Service.
BCI-I-5 - Procedures for Evaluating Plastic Jars and Covers for Motive Power Industrial Batteries.

(Copies of BCI standards may be obtained from the Battery Council International, 401 North Michigan Avenue, 24th Floor, Chicago, IL 60611.)

3. REQUIREMENTS

3.1 General. Battery components and materials shall be polypropylene (PP), high impact rubber, or polyolefin copolymers unless otherwise specified (see 6.2).

3.2 Precautionary labels. Appropriate precautionary labels shall be affixed to the batteries. Precautionary labels shall be in accordance with BCIS-10.

3.3 Cell.

3.3.1 Element. The design of the element shall ensure that the cell meets the specified electrical requirements.

3.3.2 Grids. The grids shall be made of a suitable lead alloy.

3.4 Electrolyte. The specific gravity of the electrolyte shall be within the range shown on the battery nameplate or in the manufacturer's published instructions. The electrolyte shall be in accordance with O-S-801. The values of organic impurities shall be determined in accordance with paragraph 4.4.8.1. The values of all other impurities shall be determined by any recognized test method.

3.5 Terminal markings. The battery shall be permanently marked with the polarity of the battery terminals. The marking for the positive terminal shall be "POS", "P", or "+". The marking for the negative terminal shall be "NEG", "N", or "–". Colors may also be used on the terminals to indicate polarity: red for positive and black for negative.

3.6 Cable and connectors. The battery interface cable and connector shall be of sufficient size to handle a continuous current, in amperes, equal to one fifth of the battery ampere-hour rating. The type of charging connector, position, and length of leads shall be as specified (see 6.2). Cable, cell, and terminal connectors, other than deliberately exposed lead parts, shall be protected against corrosion.

3.7 Jars. Jars shall be of single-cell or multi-cell construction. The material shall be free of rough spots, pinholes, or cracks and be capable of withstanding the tests defined in 4.4.3, 4.4.5, and 4.4.6. When subjected to the electrical breakdown test in 4.4.3, there shall be no evidence of perforation or burn through.

3.7.1 Impact resistance. The impact resistance of sample pieces cut from single-cell and multi-cell jars shall be 120 inch-pounds minimum when tested in accordance with 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

3.7.2 Sediment space. Jars shall have sufficient sediment space to minimize contact between the sediment and element that may cause a short.

3.7.3 Space above element. The space above the element shall be filled 1/2 to 3/4 inch above the tops of the separators and shall be a nominal 1/8 inch below the bottom of the filling (vent) wells.

3.7.4 Acid resistance. When tested in accordance with 4.4.6, the jars and covers shall not absorb an excessive amount of electrolyte. The material shall not be in compliance if the increase in weight exceeds 1.5 percent, if visible blisters or cracks are found, or if one or more dimensions have increased by 2 percent or more.

3.8 Cell covers. The cell cover shall be compatible with the jar. The design of the cover shall prevent distortion of the cover and leakage of electrolyte when the cell is operated in accordance with the manufacturer's instructions.

3.8.1 Vent caps. Suitable baffling shall be provided in the cell or vent to avoid spraying and minimize loss of electrolyte while permitting gas to escape during charging, discharging, or open circuit.

3.8.2 Seal. The joint between the cover and jar shall be gas-tight and shall effectively seal against leakage of electrolyte. An unbroken seal between the cover and jar shall be maintained throughout a temperature range of 0 ºF (-18 ºC) to 140 ºF (60 ºC).

3.9 Trays. Unless otherwise specified in the acquisition order (see 6.2), trays shall be of steel construction. All surfaces of metal covers and lifting means and all exposed vertical surfaces of metal trays shall be protected against exposure to sulfuric acid.

3.10 Assembly.

3.10.1 Cells. Cells shall be furnished as single cells, trays of cells, or complete batteries, as specified in the acquisition order (see 6.2).

3.10.2 Batteries. Unless otherwise specified in the acquisition order (see 6.2), the battery shall be of standard groupings, dimensions, and connections and include a charging plug, receptacle, or suitable terminal connection. Dimensions of batteries shall not vary more than 1/10 inch from the specified values. Cell connectors shall be burned (fused together) or bolted to the terminals and be adequate to carry the rated current.

3.11 Weight. The actual weight of a charged-and-wet ready for service trayed battery shall be permanently stamped into the wall of the tray near one of the lifting points in the upper case letters as "SERVICE WEIGHT - ___LBS". This weight shall be within ±5 percent of the manufacturer's published catalog value.

3.12 Charge condition. Batteries shall be shipped either charged-and-wet or charged-and-dry as specified in the acquisition order (see 6.2). When the purchaser does not specify the delivery condition, batteries shall be shipped charged-and-wet and ready for service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

3.12.1 Charged-and-wet. Charged-and-wet batteries shall be shipped filled with electrolyte of proper specific gravity, fully charged, and ready for use.

3.12.2 Charged-and-dry. The charged-and-dry delivery condition is intended for batteries that will be placed in temporary or extended storage prior to use. The element in charged-and-dry batteries shall be dry. To prevent deterioration of the element due to exposure to moisture, charged-and-dry battery vents shall be sealed. When charged-and-dry batteries are filled with electrolyte according to the manufacturer's instructions and operated as specified in 4.4.6, the batteries shall deliver 90 percent of their rated capacity at or before the third cycle.

3.13 Markings. Each batteries shall be legibly and permanently marked with the following:

a. Name or trademark of the manufacturer.
b. Manufacturer's battery part number or other identification designation.
c. battery voltage.
d. Six-hour capacity rating in ampere-hours.
e. Month and year of manufacture.
f. Service weight of trayed battery.
g. Specific gravity of electrolyte in fully charged condition.
h. Precautionary and instruction labels.

3.14 Instructions. Each battery shall have a set of instructions for the proper installation and maintenance of the battery, including cycling characteristics. Pertinent data shall identify the battery with relation to size and manufacturer. The instructions shall be securely fastened and protected from mechanical and chemical damage.

3.15 Performance.

3.15.1 Rated capacity. After conditioning in accordance with 4.4.2, the batteries shall have not less than 90 percent of their rated ampere-hour capacity at the end of the third cycle when tested in accordance with 4.4.6. The rated capacity shall not exceed the manufacturer's commercial rating. The number of cycles necessary to deliver 100 percent of the manufacturer's rated capacity shall not exceed 10.

3.15.2 Cycle life. Unless otherwise specified (see 6.2), the cell or battery shall complete a minimum of 1,500 cycles when tested in accordance with 4.5. The cells or batteries tested shall be representative of the range of capacities of the cells or batteries produced.

3.15.3 Retention of charge. Single cells or cells of a battery shall not show a drop in specific gravity greater than 0.015 units in 14 days when tested in accordance with 4.4.7.

3.16 Workmanship. The active material shall be free from breaks and loose parts. The grids shall be free of open sections, cold pouring, cracks, and blowholes. Breaks are permitted in those parts of the grid used to retain the active material only when the break does not impair the strength of the grid or permit the active material to become separated from the plate. The sections of the element involving lead burning shall be free from defective bonds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the acquisition order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the acquisition order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the tests in 4.4.

4.2.1 Lot. Lot size shall be as specified (see 6.2).

4.2.2 Sampling for inspection. Unless otherwise specified (see 6.2), the manufacturer shall select the type of sampling plan (attribute, variable, or continuous) in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

4.2.3 Major defects. The major defects for inspection purposes are listed in table IV.

TABLE IV. Major defects.

Defect	Requirement
Element	3.3.1
Grids	3.3.2
Electrolyte	3.4
Terminal markings	3.5
Connectors	3.6
Jars	3.7, 3.7.1, 3.7.2, 3.7.3, and 3.7.4
Cell covers	3.8, 3.8.1, and 3.8.2
Trays	3.9
Assembly	3.10.1 and 3.10.2
Weights	3.11
Charge condition	3.12, 3.12.1, and 3.12.2
Markings	3.13
Workmanship	3.16
Dimensions	As specified in the acquisition order

4.3 Sampling for tests. Unless otherwise specified (see 6.2), the manufacturer shall select the type of sampling plan (attribute, variable, or continuous) in accordance with MIL-STD-1916 (see 4.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Verification. Verification of tests shall be determined by review of test records and data. Results of prior testing of cells or batteries selected in accordance with 4.3 and tested in accordance with 4.4.6, capacity test; 4.4.7, retention of charge test; and 4.5, cycle life shall be acceptable to demonstrate conformance of batteries being furnished on current acquisition orders.

4.4.2 Conditioning of samples. If supplied in a charged-and-wet condition, the cell or battery may be brought to a fully charged state before discharging. If supplied in a charge-and-dry condition, the battery shall be initially conditioned in accordance with the manufacturer's instructions.

4.4.3 Electrical breakdown test. To determine conformance with 3.7, jars shall be tested in accordance with BCI-I-5. The voltage applied shall be 100 volts per mil of wall thickness.

4.4.4 Impact resistance. To determine conformance with 3.7.1, samples from jars shall be tested in accordance with BCI-I-5. The area of impact shall be the geometric center of the test specimen.

4.4.5 Acid-resistance test. To determine conformance to 3.7.4, jars and covers shall be tested in accordance with BCI-I-5.

4.4.6 Capacity test. To determine compliance with 3.15.1, cells or batteries shall be subjected to capacity tests in accordance with BCI-I-2.

4.4.7 Retention of charge test. To determine conformance with 3.15.3, the cells or batteries shall be fully charged with the specific gravity of each cell corrected to 77 ºF (25 ºC). The cell or battery shall then be allowed to stand on open circuit in an ambient temperature of 75 ºF to 90 ºF (24 ºC to 32 ºC) for a period of 14 days. At the end of the 14-day period, the specific gravity shall be measured and corrected to 77 ºF (25 ºC).

4.4.8 Electrolyte. The impurity values specified in O-S-801 (except organic matter) shall be determined by any recognized analytical test method (see 3.4).

4.4.8.1 Organic matter. Heat 200 milliliters of electrolyte in a clean beaker until the acid begins to fume strongly. To pass the test, there shall be no perceptible charring; a slight yellow color that appears when the acid is hot but disappears on cooling should be disregarded.

4.5 Cycle life. To determine conformance with 3.15.2, the cycle shall be determined in accordance with BCI-I-3. The cells shall be cycled two times a day but not less than 10 cycles per week. Verification shall be determined by review of acceptable test records and data showing compliance with the cycle requirement.

4.6 Non-conforming product. The contactor shall withhold non-conforming products from acceptance and initiate corrective action in accordance with MIL-STD-1916.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the acquisition order (see 6.2).

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

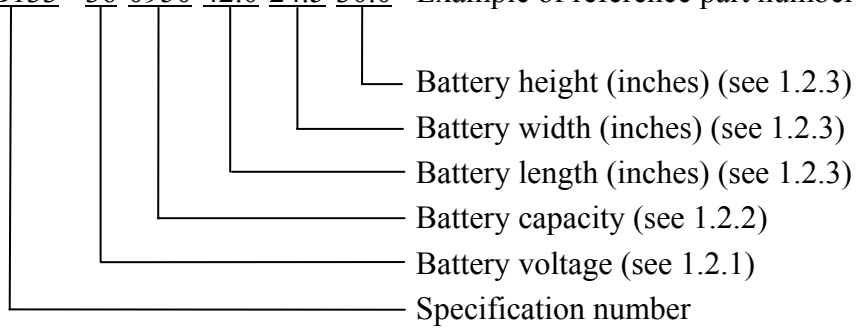
6.1 Intended use. Batteries covered by this specification are intended for use in electric fork trucks, electric tractors, and other comparable motive power applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision, and date of this specification.
b. Voltage (see 1.2.1).
c. Ampere-hour capacity (see 1.2.2).
d. Maximum overall dimensions (L x W x H) (see 1.2.3).
e. Battery components, if other than specified (see 3.1).
f. Type of charging connector, position, and length of leads (see 3.6).
g. Tray material, if other than specified (see 3.9).
h. Cells (single, tray, or complete battery) (see 3.10.1).
i. Battery grouping, if other than standard (see 3.10.2).
j. Charge condition for shipping (see 3.12).
k. Cycle life, if other than specified (see 3.15.2).
l. Inspection responsibility, if other than specified (see 4.1).
m. Lot size (see 4.2.1).
n. Inspection sampling, if other than specified (see 4.2.2).
o. Test sampling, if other than specified (see 4.3).
p. Packaging (see 5.1).

6.3 Part or identification number. The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

WB133 - 36-0950-42.0-24.5-30.0 Example of reference part number:



WB133-36-0950-42.0-24.5-30.0 identifies a battery rated at 36 volts, having a 950 ampere-hour capacity, 42.0 inches long, 24.5 inches wide, and 30 inches high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

electric
fork trucks
lead
sulfuric acid
tractors

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS2

(Project 6140-0922)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.